

Serial Interface

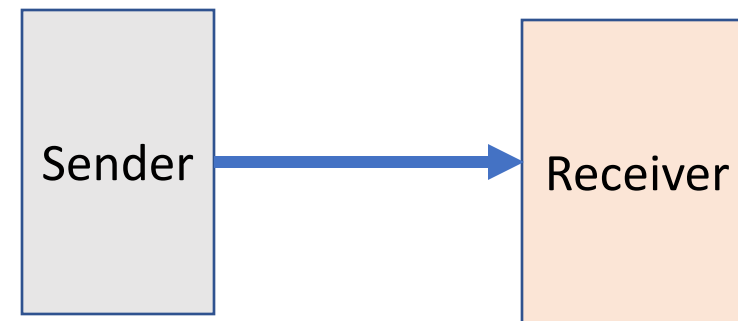
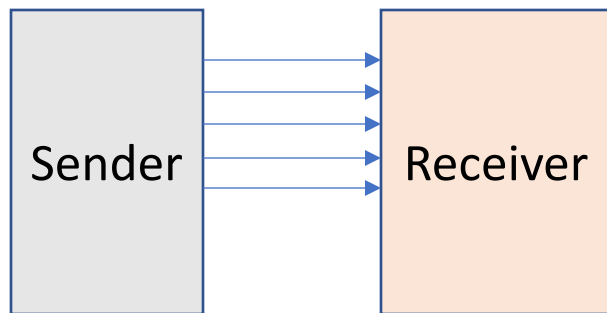
I2C and SPI

Topics

- Serial vs Parallel Interface
- SPI Serial Interface
- I2C Serial Interface

Introduction

- We need communication protocols to interface sensors or actuators with microcontrollers (or computers).
- There are two types of communications:
 - Parallel
 - Serial
- Parallel communications offer higher transmission rate but requires more lines that serial communications.



Communication Modes

One-way: data sent in one direction

Half-duplex: two way transfer, but only in one direction at a time.

Example: I2C

Full duplex: two way transfer, data sent and received simultaneously.

Example: RS-232 and SPI

Synchronous vs. Asynchronous

- In asynchronous communication data is transmitted without the use of an external clock.

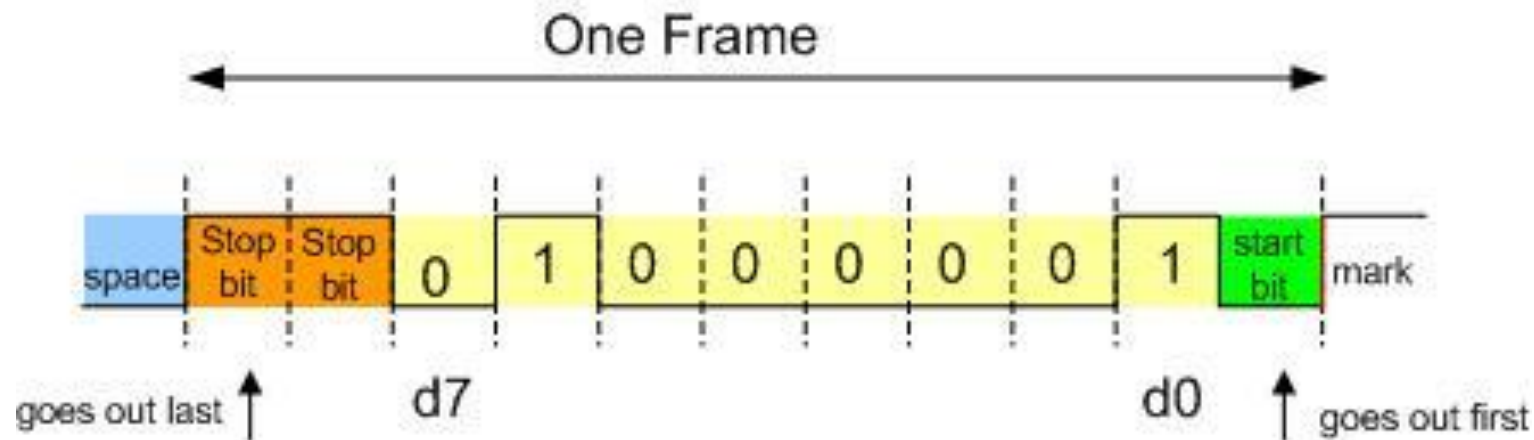
Example: RS-232 and USB

- In synchronous communication, both the transmitter and receiver share a common clock.

Example: SPI and I2C

UART Data Frame

Example: Framing ASCII "A" (0x41)

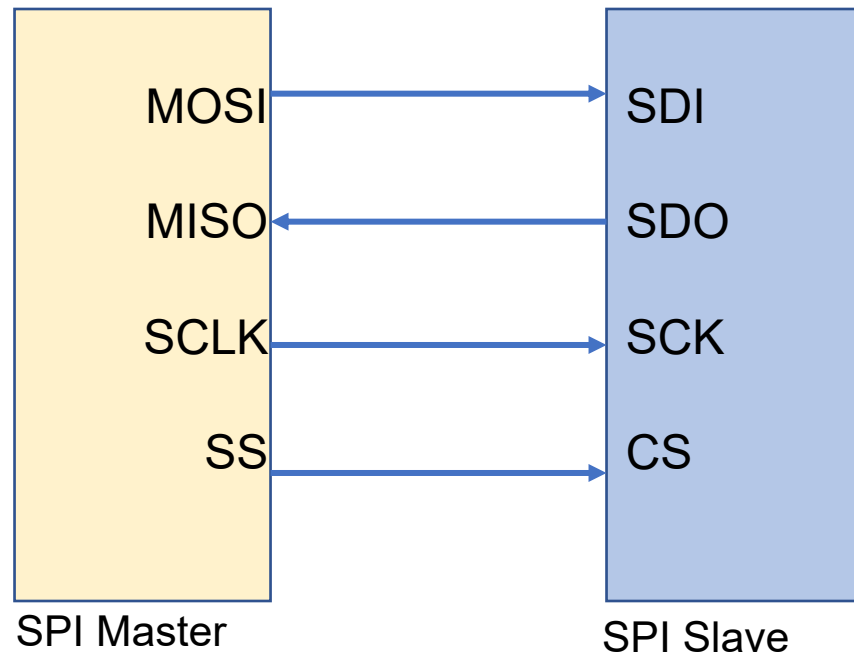


SPI Serial Interface

SPI Interface

- SPI is a 4-wire serial communication interface.
- Used by many microprocessors and microcontrollers to connect to peripheral devices such as memory modules, ADC, DAC and sensors.
- Speed up to 10 Mbps.
- Due to high speed rate, the bus lines cannot be too long and therefore SPI is only used on the PCB.
- **Full duplex** and **synchronous** mode.

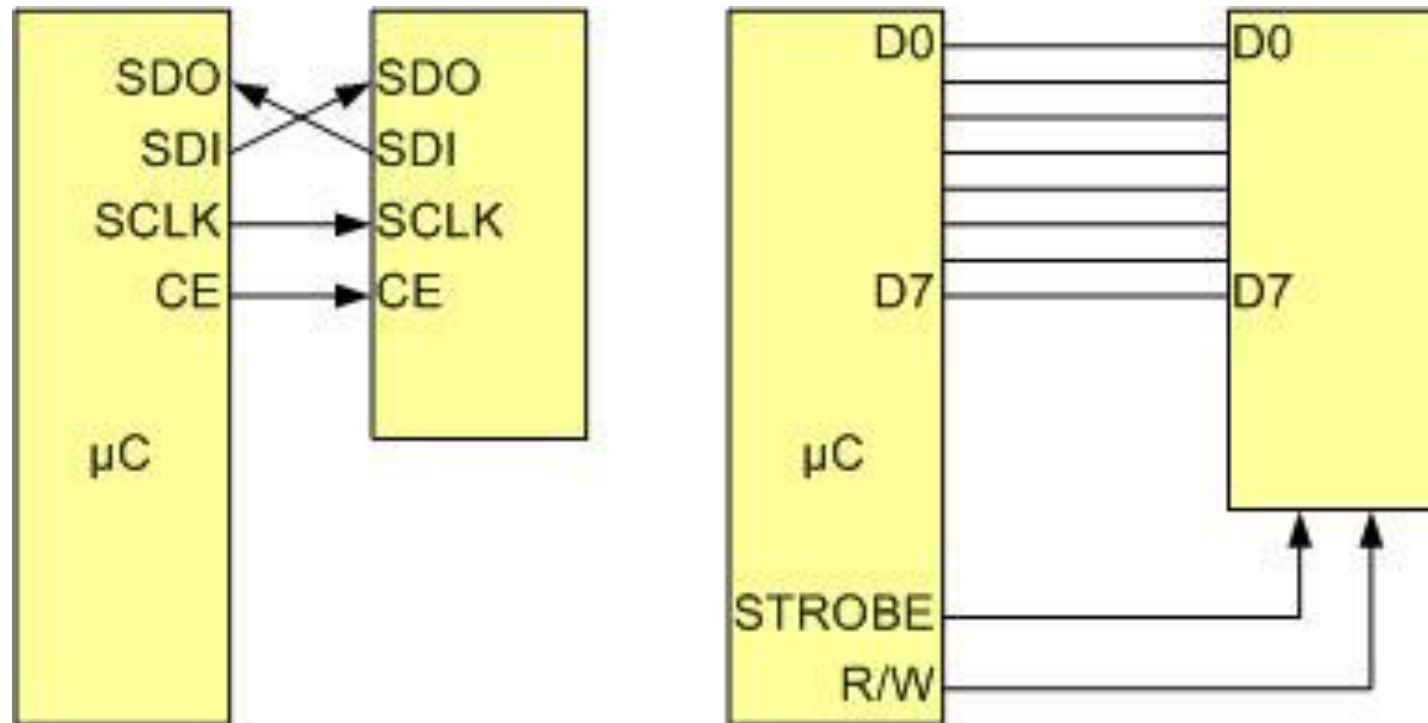
SPI connection lines



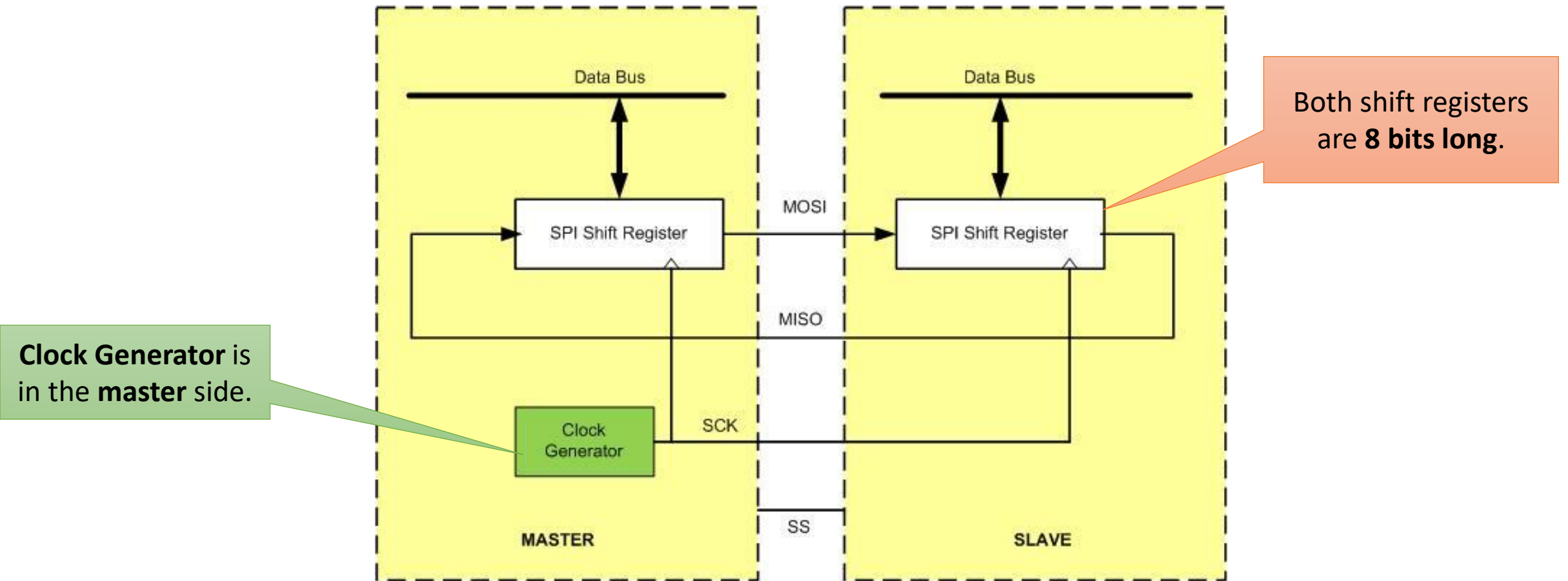
An SPI protocol has 4 wires:

- MOSI (Master Out Slave In)
- MISO (Master In Slave Out)
- SCLK or SCK (Serial Clock)
- SS (Slave select from master to chip select of slave)

SPI Bus vs. Traditional Parallel Bus Connection to Microcontroller



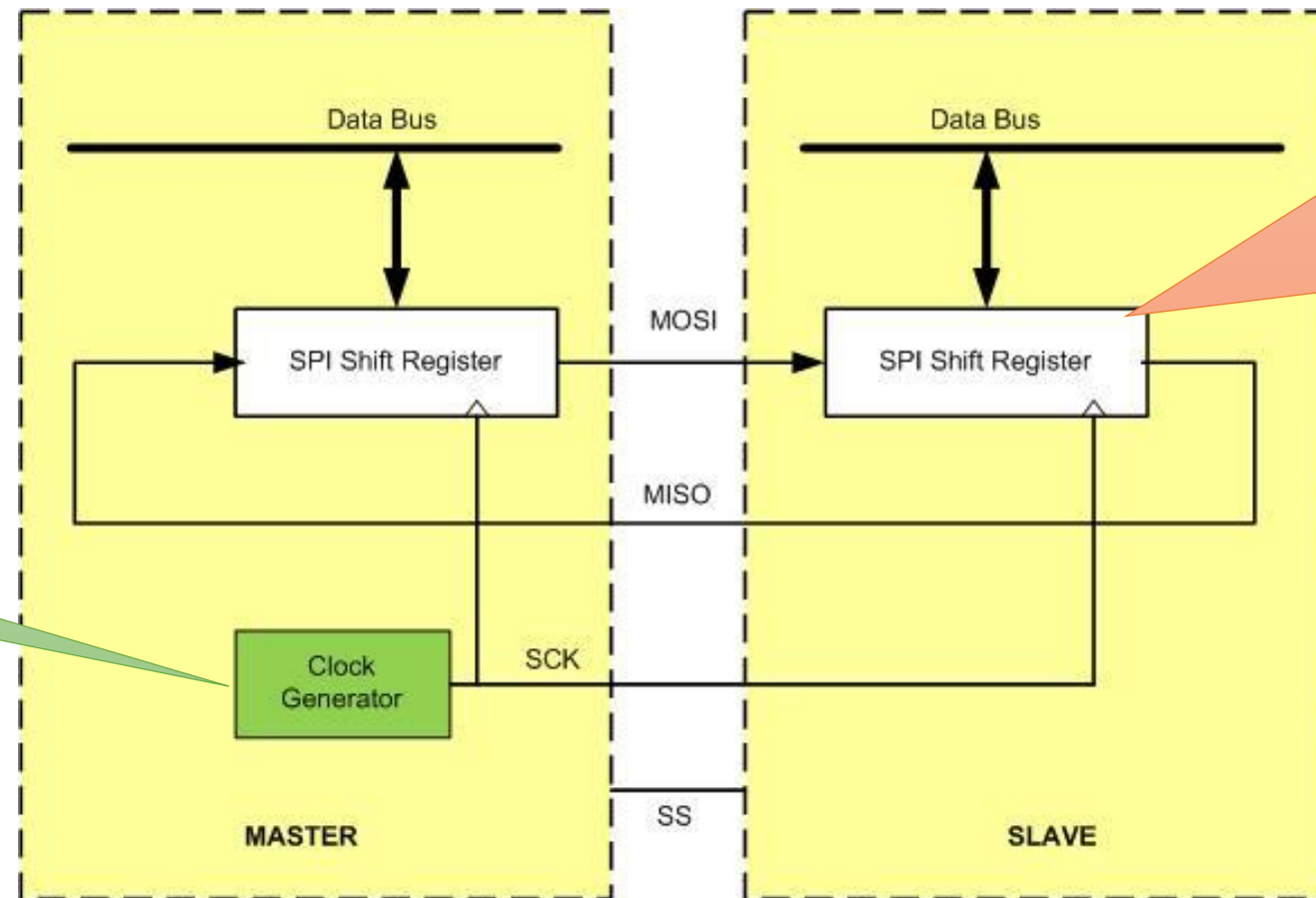
SPI Architecture





What does it mean that shift register is **8-bit long**?

SPI Architecture



Clock Generator is in the **master** side.

- Both shift registers are **8 bits long**.
- It means that after 8 clock pulses, the content of the two shift registers are interchanged.

Clock Polarity and Phase in SPI Device

- As we mentioned before in UART communication, transmitter and receiver must agree on a clock frequency (baud rate).
- In SPI communication, both master and slave uses the same clock , generated by master, but the master and slave(s) must agree on clock polarity and phase with respect to data.
- Terminologies:
 - CPOL (clock polarity)
 - CPHA (clock phase)

SPI Clock Polarity and phase

Clock Polarity	Clock Phase	Data read and change time	SPI Mode
0	0	read on falling edge, changed on a rising edge	0
0	1	read on rising edge, changed on a falling edge	1
1	0	read on rising edge, changed on a falling edge	2
1	1	read on falling edge, changed on a rising edge	3

SPI Programming in TI ARM Tiva

- The TI ARM Tiva comes with four on-chip SPI modules.
- TI literature uses the SSI (Synchronous Serial Interface) name since it supports all three protocols (SPI, MICROWIRE, and TI) synchronous serial interface.
- We examine the SPI feature of SSI module.
- The SSI modules are located at the following base address:

SSI Module	Pins
SSI0	PA0 - PA5
SSI1	PF0, PF1, PF2, PF3 or PD0 - PD3
SSI2	PB4 - PB7
SSI3	PD0 - PD3

SPI Registers in TI ARM Tiva

Register Name	Offset	Description
SSI_CR0	0x000	Control Register 0 (Configures frame format, data size, clock polarity, etc.)
SSI_CR1	0x004	Control Register 1 (Enables SSI, configures master/slave mode, loopback mode)
SSI_DR	0x008	Data Register (Read/write data for SPI communication)
SSI_SR	0x00C	Status Register (Shows if TX/RX buffers are full/empty)
SSI_CPSR	0x010	Clock Prescale Register (Sets the clock rate for SPI communication)
SSI_IM	0x014	Interrupt Mask Register (Enables or disables SPI interrupts)
SSI_RIS	0x018	Raw Interrupt Status Register (Shows unmasked interrupt status)
SSI_MIS	0x01C	Masked Interrupt Status Register (Shows masked interrupt status)
SSI_ICR	0x020	Interrupt Clear Register (Clears interrupt flags)
SSI_DMACTL	0x024	DMA Control Register (Enables DMA for TX/RX)
SSI_CC	0xFC8	Clock Configuration Register (Configures clock source for SPI)

Credit: Table generated with the help of GenAI

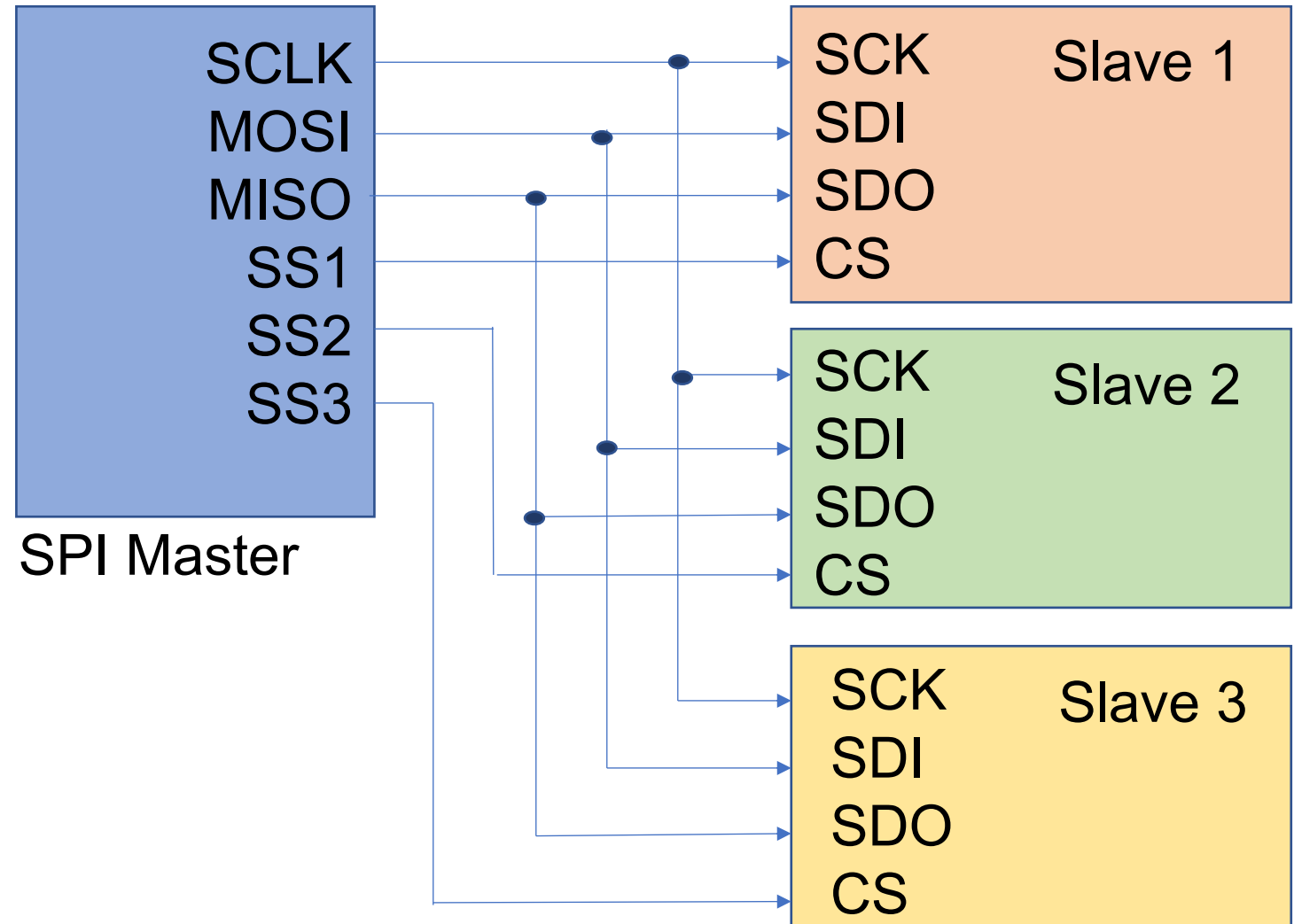
Configuring SSI Module for SPI Protocol

1. Enable the clock to SSI module using RCGCSSI.
2. Disable the SSI via bit1 of SSI_CR1 register before initializing.
3. Set the bit rate with the SSI_CPSR prescaler and SSI_CR0 control registers.
4. Select the SPI mode, phase, polarity, and data size in SSI_CR0 control register.
5. Select the master mod in SSI_SCR1 register.
6. Enable the SSI via bit1 of SSI_CR1 register.
7. Assert slave select signal.
8. Wait until the TNF flag (transmit FIFO not full) is SSI_SR(status register) goes high, then load a byte of data into SSI_DR (data register) to be transmitted.
9. Repeat step 8 above until all the data are in FIFO.
10. Wait until transmit is complete (transmit FIFO empty and SSI not busy).
11. Deassert the slave select signal.

Example

- Please refer to source code “Program 8-1” on BB for an example of sending A-Z characters via SPI1.

SPI is a Multi-slave, Single-Master Protocol

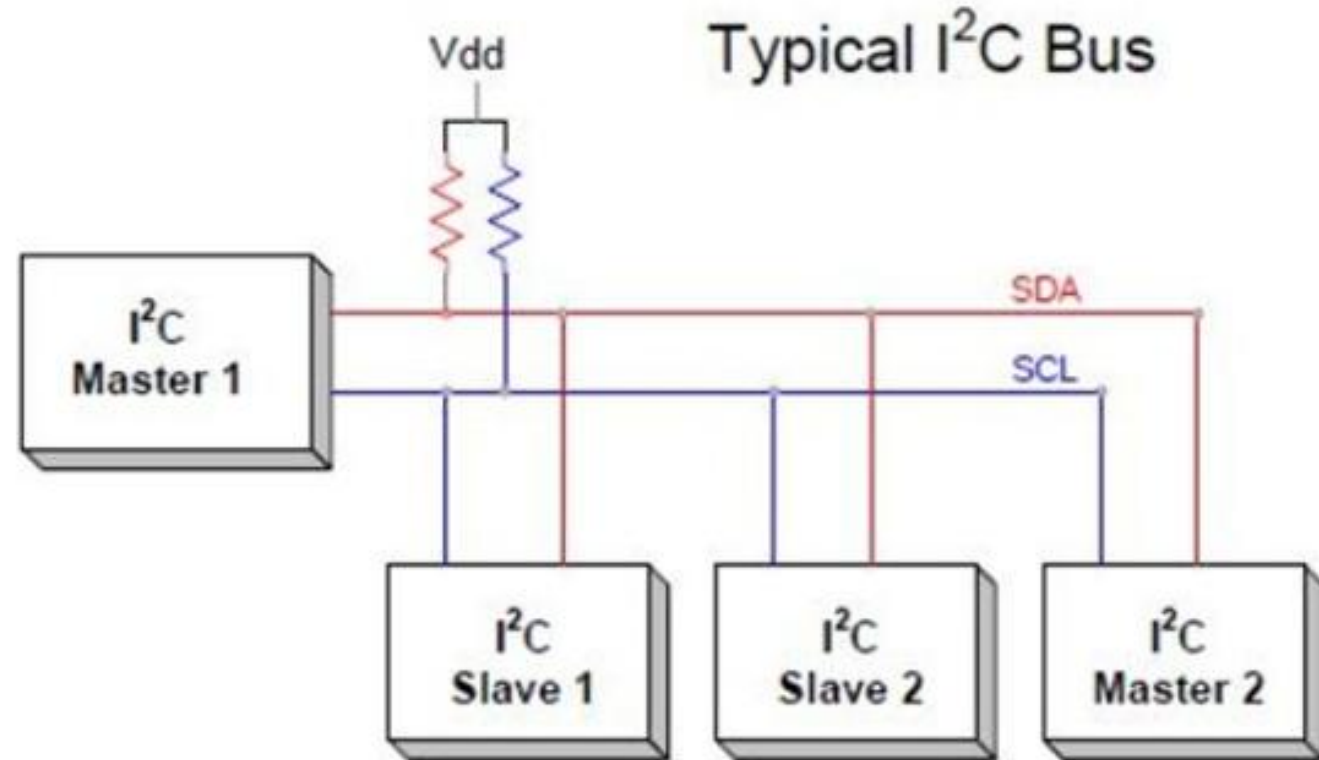


I2C Serial Communication

I2C Interface

- I2C (Inter Integrated Circuits) bus is a **bi-directional** 2-wire bus.
- I2C has two bi-directional lines:
 - SDA** – Serial Data Line
 - SCL** – Serial Clock Line
- It is a serial communication widely used for serial communication between integrated circuits on the same circuit board and to attach low-speed peripheral devices, sensors and components to embedded systems.

Typical I2C Bus



Both lines require **pull-up resistors** (typically 4.7K) since devices only pull the line low (open-drain configuration).

I2C Nodes

- In I2C Protocol, more than 100 devices can share an I2C bus.
- Each device is called a node.
- I2C bus is **multi-master**. It means any of the devices on the bus can initiate data transfer.

Master -- The device that starts the communication.

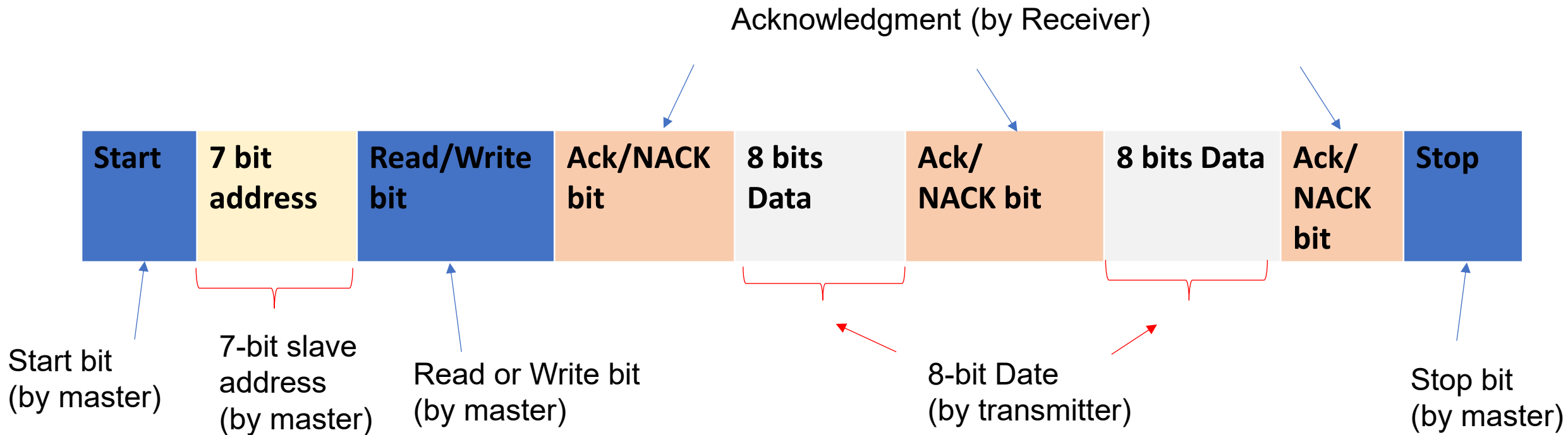
Slave -- The device that is addressed by master.

- The **clock** is generated by master.
- I2C uses **7 bit addresses**. This means that you can connect up to 128 devices on the I2C bus, since 7 bit numbers can be from 0 to 127.

I2C Communication Process

1. **Start Condition:** Master pulls SDA low while SCL is high.
2. **Addressing:** Master sends the 7-bit address + R/W bit.
3. **ACK/NACK:** Slave acknowledges (ACK) or does not acknowledge (NACK).
4. **Data Transfer:** Master and slave exchange data, 8-bit at a time.
5. **Stop Condition:** Master releases SDA and SCL.

I2C Communication Protocol



I2C Programming in TI ARM Tiva

- IVA C (TM4C123G) has **four I2C modules** (I2C0 to I2C3) for communication.
- Each module has specific SCL and SDA pins assigned in the GPIO peripheral.

I2C Module	SCL Pin	SDA Pin
I2C0	PB2	PB3
I2C1	PA6	PA7
I2C2	PE4	PE5
I2C3	PD0	PD1

I2C Register Overview

Register Name	Offset	Description
I2CMCR	0x020	I2C Master Configuration
I2CMTPR	0x00C	I2C Master Timer Period (Sets SCL clock speed)
I2CMCS	0x004	I2C Master Control/Status
I2CMDR	0x008	I2C Master Data
I2CMRIS	0x00C	I2C Master Raw Interrupt Status
I2CMIMR	0x010	I2C Master Interrupt Mask
I2CMCR2	0x038	I2C Master Configuration 2
I2CMSA	0x000	I2C Master S ⁷ Address

Credit: Table generated with GenAI

Configuring I2C on TIVA C

- 1.Enable the I2C module
- 2.Configure the GPIO pins for I2C functionality
- 3.Initialize the I2C master or slave
- 4.Send and receive data

Reference

1. Mazidi, M., Chen, S., Naimi, S., Naimi, S., Chen, S. (2016). TI TIVA ARM Programming For Embedded Systems, 1st Edition. MicroDigitalEd. ISBN-13: 978-0997925920
2. Valvano, J., (2019). Embedded Systems: Introduction to ARM Cortex-M Microcontrollers, 5th edition, 6th printing